

EVALUATION-2 CHEAT SHEET

Ratio & Partnership · Speed / Time / Distance & Trains · Blood Relations · Direction Sense · Syllogism

1. RATIO & PARTNERSHIP

Core Formulas

$$a : b = a / b$$

$$\text{If } a:b = c:d \rightarrow a \times d = b \times c \text{ (cross multiply)}$$

$$\text{Compounded ratio: } (a:b) \times (c:d) = ac : bd$$

$$\text{Duplicate: } a:b \rightarrow a^2 : b^2 \mid \text{Sub-dup: } \sqrt{a} : \sqrt{b}$$

$$A:B = a:b, B:C = c:d \rightarrow A:B:C = ac : bc : bd$$

$$\text{Share formula: } A = [x/(x+y+z)] \times \text{Total}$$

$$\text{Componendo: } a/b = c/d \rightarrow (a+b)/b = (c+d)/d$$

$$\text{Dividendo: } a/b = c/d \rightarrow (a-b)/b = (c-d)/d$$

$$\text{Comp-Div: } (a+b)/(a-b) = (c+d)/(c-d)$$

Partnership

$$\text{Simple: Profit ratio} = \text{Capital ratio (same time)}$$

$$\text{Compound: } A:B \text{ profit} = (a \times t_1) : (b \times t_2)$$

Working partner: deduct salary first, then split by capital

Variation

$$\text{Direct: } a_1/b_1 = a_2/b_2$$

$$\text{Inverse: } a_1 \times b_1 = a_2 \times b_2$$

2. SPEED, TIME & DISTANCE & TRAINS

Core Formula

$$D = S \times T \rightarrow S = D/T \rightarrow T = D/S$$

$$\text{km/h} \rightarrow \text{m/s} : \text{multiply by } 5/18$$

$$\text{m/s} \rightarrow \text{km/h} : \text{multiply by } 18/5$$

Quick ref: $72 \text{ km/h} = 20 \text{ m/s}$ | $36 \text{ km/h} = 10 \text{ m/s}$ | $18 \text{ km/h} = 5 \text{ m/s}$

Average Speed

$$\text{Same TIME for both legs: Avg} = (U + V) / 2$$

$$\text{Same DISTANCE both legs: Avg} = 2UV / (U + V) \triangle$$

Use this for return journeys!

Relative Speed

Direction	Relative Speed
Same direction	$ S_1 - S_2 $
Opposite direction	$S_1 + S_2$

Trains

Scenario	Distance covered
Cross a pole / person	Length of train (L)
Cross a platform / bridge	$L(\text{train}) + L(\text{platform})$
Two trains cross each other	$L_1 + L_2$
Train pass walking person	$L(\text{train})$ only

$$\text{Time} = \text{Distance} \div \text{Relative Speed (always!)}$$

3. BLOOD RELATIONS

Family Tree Method

+ = Male | - = Female

Grandparents (top) → Parents (middle) → Children (bottom)

Horizontal line between couple = married

Downward arrow = parent → child

Horizontal line between two people = siblings

Coded relations: decode each symbol first, THEN draw tree

Step-by-step Strategy

1. Assign +/− as soon as gender is known
2. Connect couple side-by-side with a horizontal line
3. Drop downward arrow to each child
4. Trace path from one person to another
5. Name each hop: parent / child / sibling / spouse

Key Relationships

Relation	Who they are
Uncle / Aunt	Parent's sibling
Nephew / Niece	Sibling's child
Cousin	Uncle / Aunt's child
Father-in-law	Spouse's father
Mother-in-law	Spouse's mother
Brother-in-law	Spouse's brother OR sister's husband
Sister-in-law	Spouse's sister OR brother's wife
Maternal uncle	Mother's brother
Paternal uncle	Father's brother

4. DIRECTION SENSE

Turn Rules

Facing	Left turn →	Right turn →
North	West	East
South	East	West
East	North	South
West	South	North
North-East	North-West	South-East
South-West	South-East	North-West

90° CW = right turn | 90° ACW = left turn | 180° = reverse

Shortest Distance

$d = \sqrt{(\text{Net EW}^2 + \text{Net NS}^2)}$ (Pythagoras)

Net EW = total East − total West

Net NS = total North − total South

Pythagorean Triplets (memorise!)

a	b	Hyp
3	4	5
5	12	13
7	24	25
8	15	17
6	8	10
9	40	41

Shadow Rules

Morning (sun in East) → shadow falls WEST

Evening (sun in West) → shadow falls EAST

Noon → no shadow (sun overhead)

5. SYLLOGISM

Statement Types

Type	Code	Form	Venn Diagram
Universal Positive	A	All S are P	Circle S fully inside circle P
Universal Negative	E	No S is P	Circles S and P completely separate
Particular Positive	I	Some S are P	Circles S and P partially overlap
Particular Negative	O	Some S are not P	Part of S is outside circle P

Valid Conversions

Original	Converts to	Note
A: All S → P	I: Some P → S	A weakens to I
E: No S → P	E: No P → S	Fully reversible
I: Some S → P	I: Some P → S	Fully reversible
O: Some S not P	No valid conversion	O cannot convert

Conclusion Rules (two premises, middle term M)

P1 + P2	Conclusion
A + A	A (All S are P)
A + E	E (No S is P)
A + I	I (Some S are P)
A + O	O (Some S not P)
E + A	O* (Some P not S)
E + I	O* (Some P not S)
I + A	I (Some S are P)
I + E	O (Some S not P)
O + any	No definite conclusion
I + I	No definite conclusion
E + E	No definite conclusion

* Verify with Venn diagram — middle term must be distributed

Thumb Rules

Two NEGATIVE premises → No conclusion

Two PARTICULAR premises (I or O) → No conclusion

One negative premise → Conclusion must be negative

One particular premise → Conclusion must be particular

Two universal premises → Conclusion can be universal

Complementary Pairs (Either - Or conclusions)

When NEITHER conclusion follows individually → check if they form a complementary pair for the SAME pair of terms:

Pair Type	Example
I + O	Some A are B + Some A are not B
A + O	All A are B + Some A are not B
I + E	Some A are B + No A is B

If YES → answer is "Either I or II follows". Both conclusions must use THE SAME pair of terms.

Step-by-Step Approach

1. Identify the middle term (M) shared between both premises
2. Determine types: A / E / I / O for each premise
3. Apply conclusion rule from the table above
4. Check each given conclusion — does it match or follow from the derived conclusion?
5. If nothing follows → test complementary pair condition

MASTER QUICK REFERENCE — All 5 Topics

Topic	Key Formula / Rule	Watch Out For
Ratio	$A:B:C = ac : bc : bd$ $\text{Share} = \left[\frac{x}{x+y+z} \right] \times \text{Total}$ $\text{Comp-Div: } (a+b)/(a-b) = (c+d)/(c-d)$	LCM method to merge 2-term ratios into 3-term
Partnership	$\text{Profit ratio} = \text{Capital} \times \text{Time}$ Working partner: salary deducted first	Different joining months = compound partnership
Speed / Dist	$D = S \times T$ $72 \text{ km/h} = 20 \text{ m/s } (\times 5/18)$ $\text{Same dist avg} = 2UV/(U+V)$	NEVER use $(U+V)/2$ for same-distance return trips
Trains	$\text{Time} = (L1+L2) / \text{Relative Speed}$ $\text{Same dir: } S1-S2 \quad \quad \text{Opp dir: } S1+S2$	Always add BOTH train lengths; convert speed to m/s
Blood Relations	Draw family tree, assign + / - Trace path hop by hop	Coded relations — decode symbol table before drawing
Direction	$d = \sqrt{EW^2 + NS^2}$ $\text{Left turn: } N \rightarrow W \rightarrow S \rightarrow E \quad \quad \text{Right: } N \rightarrow E \rightarrow S \rightarrow W$	Shadow: morning = shadow West; evening = shadow East
Syllogism	$A+A=A, A+E=E, A+I=I, E+I=O, I+A=I$ Two negatives / two particulars = no conclusion	Complementary pair ($I+O, A+O, I+E$) → Either-Or and Not Both

Eval-2 Cheat Sheet | 25 Questions | 45 Minutes | All formulas, rules, and triplets in one place